Welcome To this Session on Cloud Computing With AWS

AGENDA

Module I- INTRODUCTION TO AWS

Module II- IAM (Identity access management)

Module III - EC2 (elastic cloud compute)

Module IV – High availability and scalability

Module V- EC2 storage

Module VI- Route 53

Module VII- Amazon s3 introduction

Module VIII- AWS monitoring and audit cloudwatch and cloudtrail

Module IX- VPC

Module I- INTRODUCTION TO AWS

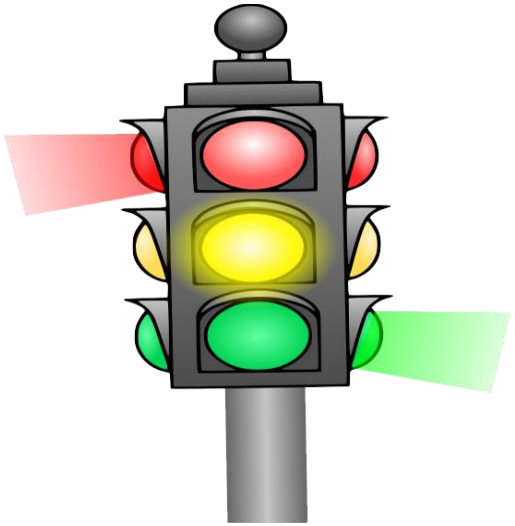
- What happened before the cloud technology emerged?

Before the rise of cloud, what was the approach to run an application on the internet before cloud?

On Premise Setup

Buy stack of servers and others hardware components.

Maintain & update the servers



Consider websites traffic changes and accordingly

Monitoring and reporting

Recruit top security professional

Disadvantages of on-premise Setup

Maintenance of servers

Increase in expenditure

Data privacy & Security is poor.

Scalability and Flexibility

What’s cloud computing?

- What is cloud?

A cloud is data center hardware and software that the vendor uses to offer the computing resources and services.

- cloud computing

Represents both cloud & the provided services

- In the simplest terms, cloud computing is a technology where a resource is provided as a service trough the internet to a user
- Cloud computing is the on-demand delivery of compute power, database, storage, applications, and other IT resources through a cloud services platform via the Internet with pay-as-you-go pricing.

Cloud Computing

- **Who is Who…**

Cloud providers

Cloud Users & Service Providers

Service Users

"With Amazon [AWS], on Day One of launch we could scale to the world."

-Brad Jefferson, Co-Founder & CEO, Animoto

Cloud Computing Benefits

- **Trade capital expense for variable expense** – Instead of having to invest heavily in data centers and servers before you know how you're going to use them, you can pay only when you consume computing resources, and pay only for how much you consume.
- **Benefit from massive economies of scale** – By using cloud computing, you can achieve a lower variable cost than you can get on your own. Because usage from hundreds of thousands of customers is aggregated in the cloud, providers such as AWS can achieve higher economies of scale, which translates into lower pay as-you-go prices.
- **Stop guessing capacity** – Eliminate guessing on your infrastructure capacity needs. When you make a capacity decision prior to deploying an application, you often end up either sitting on expensive idle resources or dealing with limited capacity. With cloud computing, these problems go away. You can access as much or as little capacity as you need, and scale up and down as required with only a few minutes' notice.
- **Increase speed and agility** – In a cloud computing environment, new IT resources are only a click away, which means that you reduce the time to make those resources available to your developers from weeks to just minutes. This results in a dramatic increase in agility for the organization, since the cost and time it takes to experiment and develop is significantly lower.
- **Stop spending money running and maintaining data centers** – Focus on projects that differentiate your business, not the infrastructure. Cloud computing lets you focus on your own customers, rather than on the heavy lifting of racking, stacking, and powering servers.
- **Go global in minutes** – Easily deploy your application in multiple regions around the world with just a few clicks. This means you can provide lower latency and a better experience for your customers at minimal cost.

Types of Cloud Computing

Cloud computing provides developers and IT departments with the ability to focus on what matters most and avoid undifferentiated work such as procurement, maintenance, and capacity planning. As cloud computing has grown in popularity, several different models and deployment strategies have emerged to help meet specific needs of different users. Each type of cloud service and deployment method provides you with different levels of control, flexibility, and management. Understanding the differences between Infrastructure as a Service, Platform as a Service, and Software as a Service, as well as what deployment strategies you can use, can help you decide what set of services is right for your needs.

Cloud Computing Models

- **Infrastructure as a Service (IaaS)**
- Infrastructure as a Service (IaaS) contains the basic building blocks for cloud IT and typically provides access to networking features, computers (virtual or on dedicated hardware), and data storage space. IaaS provides you with the highest level of flexibility and management control over your IT resources and is most similar to existing IT resources that many IT departments and developers are familiar with today.
- **Platform as a Service (PaaS)**
- Platform as a Service (PaaS) removes the need for your organization to manage the underlying infrastructure (usually hardware and operating systems) and allows you to focus on the deployment and management of your applications. This helps you be more efficient as you don’t need to worry about resource procurement, capacity planning, software maintenance, patching, or any of the other undifferentiated heavy lifting involved in running your application.
- **Software as a Service (SaaS)**
- Software as a Service (SaaS) provides you with a completed product that is run and managed by the service provider. In most cases, people referring to Software as a Service are referring to end-user applications. With a SaaS offering you do not have to think about how the service is maintained or how the underlying infrastructure is managed; you only need to think about how you will use that particular piece of software. A common example of a SaaS application is web-based email which you can use to send and receive email without having to manage feature additions to the email product or maintain the servers and operating systems that the email program is running on.

Cloud Computing Deployment Models

- **Cloud**
- A cloud-based application is fully deployed in the cloud and all parts of the application run in the cloud. Applications in the cloud have either been created in the cloud or have been migrated from an existing infrastructure to take advantage of the benefits of cloud computing. Cloud-based applications can be built on low-level infrastructure pieces or can use higher level services that provide abstraction from the management, architecting, and scaling requirements of core infrastructure.
- **Hybrid**
- A hybrid deployment is a way to connect infrastructure and applications between cloud-based resources and existing resources that are not located in the cloud. The most common method of hybrid deployment is between the cloud and existing on-premises infrastructure to extend, and grow, an organization's infrastructure into the cloud while connecting cloud resources to the internal system. For more information on how AWS can help you with your hybrid deployment, visit our Hybrid Cloud with AWS page.
- **On-premises**
- The deployment of resources on-premises, using virtualization and resource management tools, is sometimes called the “private cloud.” On-premises deployment doesn’t provide many of the benefits of cloud computing but is sometimes sought for its ability to provide dedicated resources. In most cases this deployment model is the same as legacy IT infrastructure while using application management and virtualization technologies to try and increase resource utilization. For more information on how AWS can help, see Use case: Cloud services on-premises.

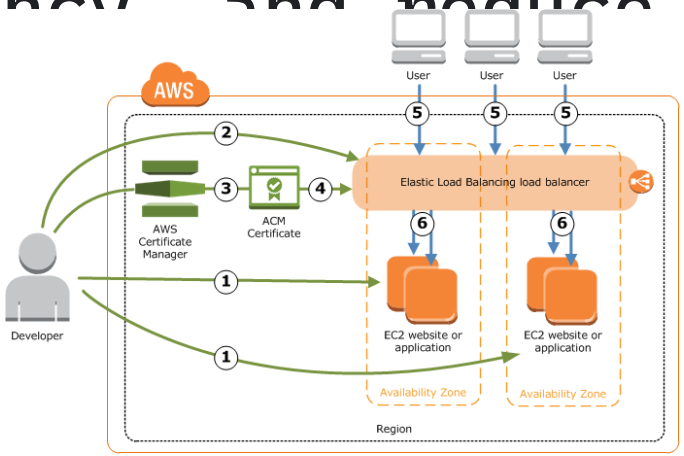
Cloud Computing Benefits

- Data privacy and security
- No maintenance worries
- Faster Data recovery
- Scale dynamically
- Reduced costs

What's AWS

- Amazon Web Services is a cloud provider which provides its users a wide range of services via internet.

Most enterprise customers across industries choose AWS as their Cloud platform. Goldman Sachs, Coca-Cola, Johnson & Johnson, Siemens, Shell, Netflix and Comcast are some of the thousands of enterprises using AWS to create new revenue streams, improve operational efficiency, and reduce business risks.



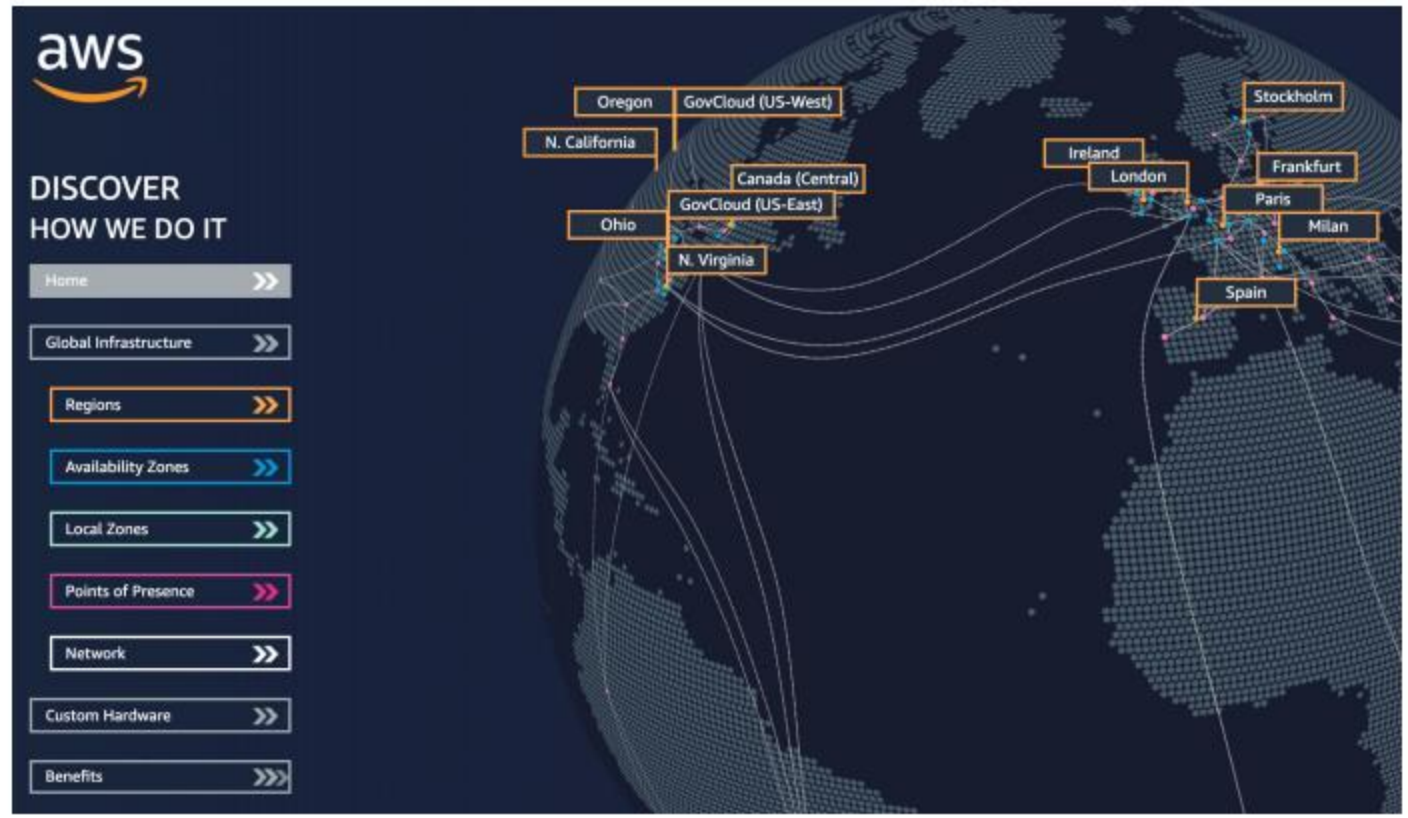
Global Infrastructure

- AWS serves over a million active customers in more than 240 countries and territories. We are steadily expanding global infrastructure to help our customers achieve lower latency and higher throughput, and to ensure that their data resides only in the AWS Region they specify. As our customers grow their businesses, AWS will continue to provide infrastructure that meets their global requirements.
- The AWS Cloud infrastructure is built around AWS Regions and Availability Zones. An AWS Region is a physical location in the world where we have multiple Availability Zones. Availability Zones consist of one or more discrete data centers, each with redundant power, networking, and connectivity, housed in separate facilities. These Availability Zones offer you the ability to operate production applications and databases that are more highly available, fault tolerant, and scalable than would be possible from a single data center. The AWS Cloud operates in 80 Availability Zones within 25 geographic Regions around the world, with announced plans for more Availability Zones and Regions.
- Each Amazon Region is designed to be completely isolated from the other Amazon Regions. This achieves the greatest possible fault tolerance and stability. Each Availability Zone is isolated, but the Availability Zones in a Region are connected through low-latency links. AWS provides you with the flexibility to place instances and store data within multiple geographic regions as well as across multiple Availability Zones within each AWS Region. Each Availability Zone is designed as an independent failure zone. This means that Availability Zones are physically separated within a typical metropolitan region and are located in lower risk flood plains (specific flood zone categorization varies by AWS Region). In addition to discrete uninterruptible power supply (UPS) and onsite backup generation facilities, data centers located in different Availability Zones are designed to be supplied by independent substations to reduce the risk of an event on the power grid impacting more than one Availability Zone. Availability Zones are all redundantly connected to multiple tier-1 transit providers.

AWS Global Infrastructure

- AWS Regions
- AWS Availability Zones
- AWS Data Centers
- AWS Edge Locations / Points of Presence

- https://infrastructure.aws/



AWS Regions

- AWS has **Regions** all around the world
- Names can be us-east-1, eu-west-3…
- A region is a **cluster of data centers**
- Most AWS services are region-scoped

https://aws.amazon.com/about-aws/global-infrastructure/

US East (N. Virginia)	us-east-1
US East (Ohio)	us-east-2
US West (N. California)	us-west-1
US West (Oregon)	us-west-2
Africa (Cape Town)	af-south-1
Asia Pacific (Hong Kong)	ap-east-1
Asia Pacific (Mumbai)	ap-south-1
Asia Pacific (Seoul)	ap-northeast-2
Asia Pacific (Singapore)	ap-southeast-1
Asia Pacific (Sydney)	ap-southeast-2
Asia Pacific (Tokyo)	ap-northeast-1
Canada (Central)	ca-central-1
Europe (Frankfurt)	eu-central-1
Europe (Ireland)	eu-west-1
Europe (London)	eu-west-2
Europe (Paris)	eu-west-3
Europe (Stockholm)	eu-north-1
Middle East (Bahrain)	me-south-1
South America (São Paulo)	sa-east-1

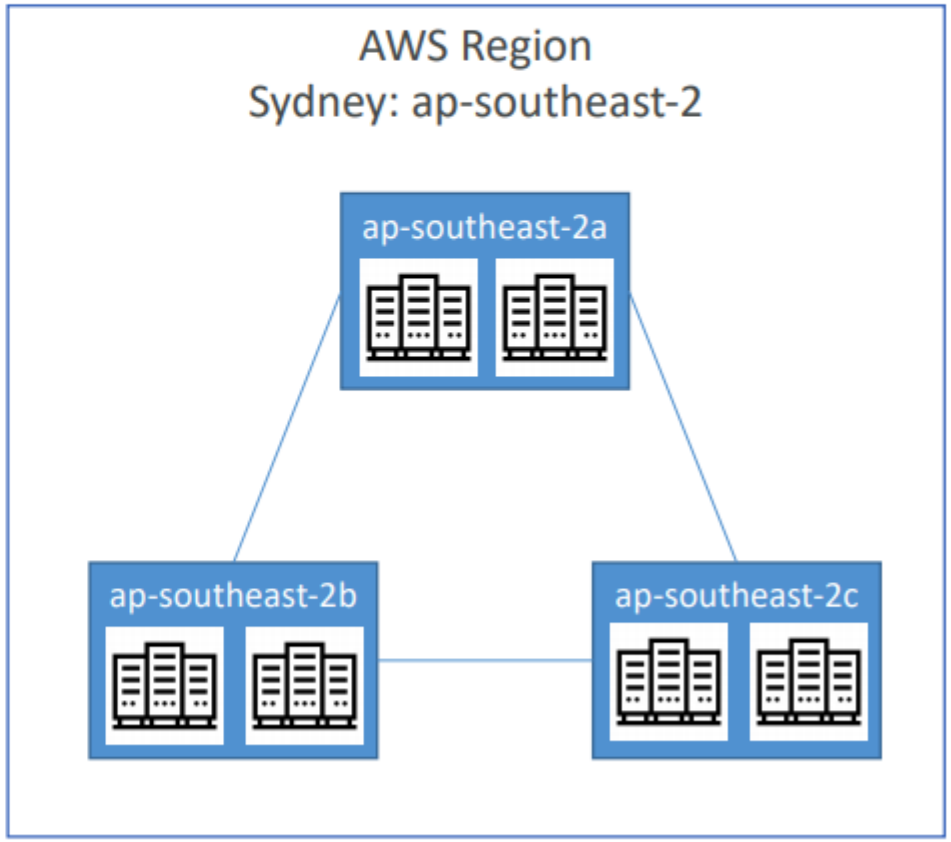
How to choose an AWS Region?

If you need to launch a new application, where should you do it?

- Compliance **with data governance and legal requirements:** data never leaves a region without your explicit permission
- Proximity **to customers:** reduced latency
- Available services **within a Region:** new services and new features aren't available in every Region
- Pricing**:** pricing varies region to region and is transparent in the service pricing page

AWS Availability Zones

- Each region has many availability zones (usually 3, min is 2, max is 6). Example:
  - ap-southeast-2a
  - ap-southeast-2b
  - ap-southeast-2c
- Each availability zone (AZ) is one or more discrete data centers with redundant power, networking, and connectivity
- They're separate from each other, so that they're isolated from disasters
- They're connected with high bandwidth, ultra-low latency networking



AWS Points of Presence (Edge Locations)

- Amazon has 216 Points of Presence (205 Edge Locations & 11 Regional Caches) in 84 cities across 42 countries
- Content is delivered to end users with lower latency

Tour of the AWS Console

- AWS has Global Services:
  - Identity and Access Management (IAM)
  - Route 53 (DNS service)
  - CloudFront (Content Delivery Network)
  - WAF (Web Application Firewall)
- Most AWS services are Region-scoped:
  - Amazon EC2 (Infrastructure as a Service)
  - Elastic Beanstalk (Platform as a Service)
  - Lambda (Function as a Service)
  - Rekognition (Software as a Service)
- Region Table: https://aws.amazon.com/about-aws/global-infrastructure/regional-product-services

AWS CORE SERVICES

Compute, Storage, Networking, and Databases

CATEGORY	SERVICE	PURPOSE
Compute	EC2	Virtual servers
Compute	Lambda	Serverless functions
Storage	S3	Object storage
Database	RDS	Relational database

Pricing Models and Cost Management

PRICING MODEL	DESCRIPTION
On-Demand	Pay per hour or second without commitment
Reserved	Discounted rates for long-term usage
Spot	Low-cost access to unused capacity

Others Cloud providers

Microsoft Azure

Google Cloud

Alibaba Cloud

IBM Cloud

vmware®

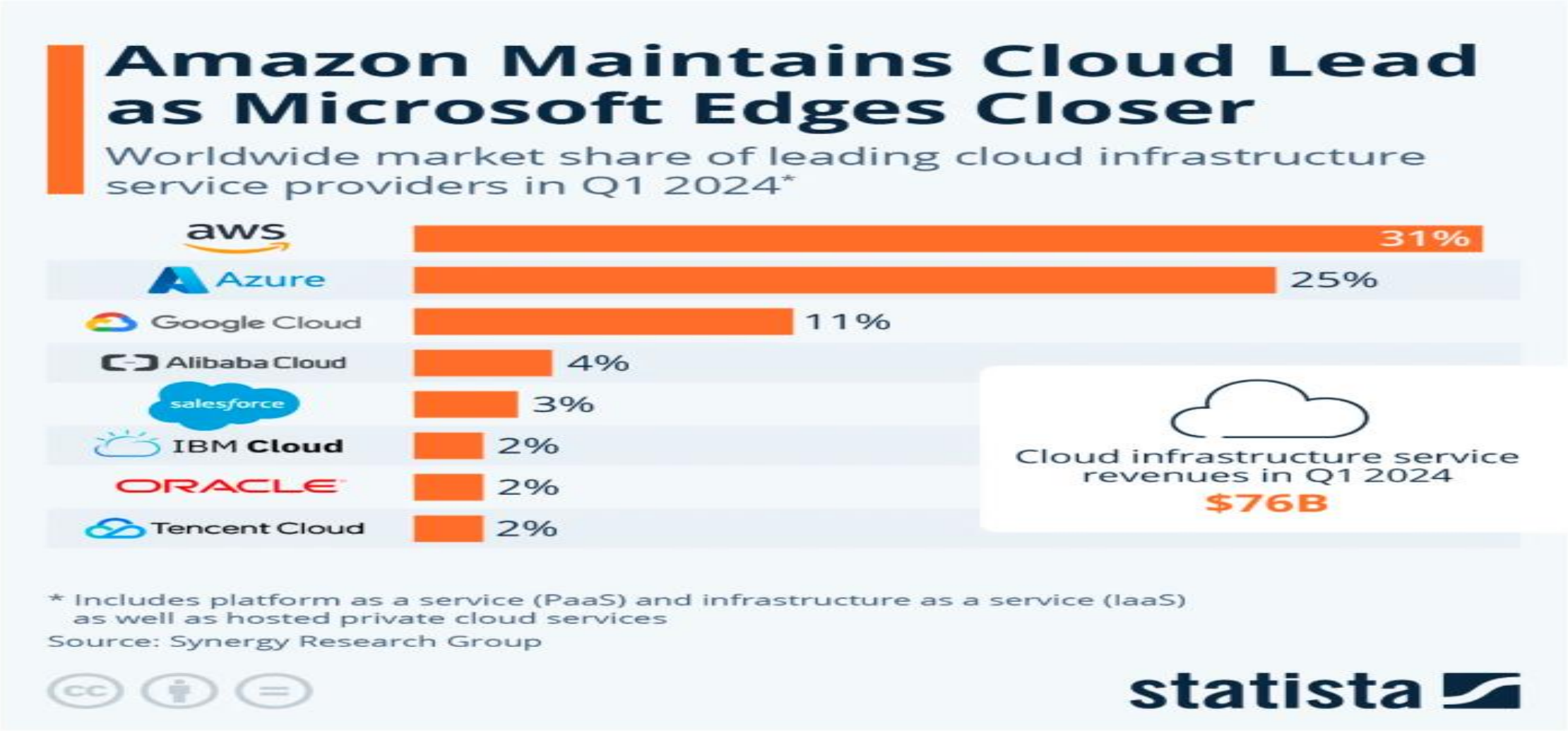
salesforce marketing cloud

Why is AWS so successful?

Amazon Web Services revolutionized the IT industry by introducing companies a new way to use servers

- Simple and per hour billing
- Amazon's Brand name
- Easy profile setup

Cloud Computing Market Share



Credit: Cloud infrastructure services market share

Amazon web services Job strands

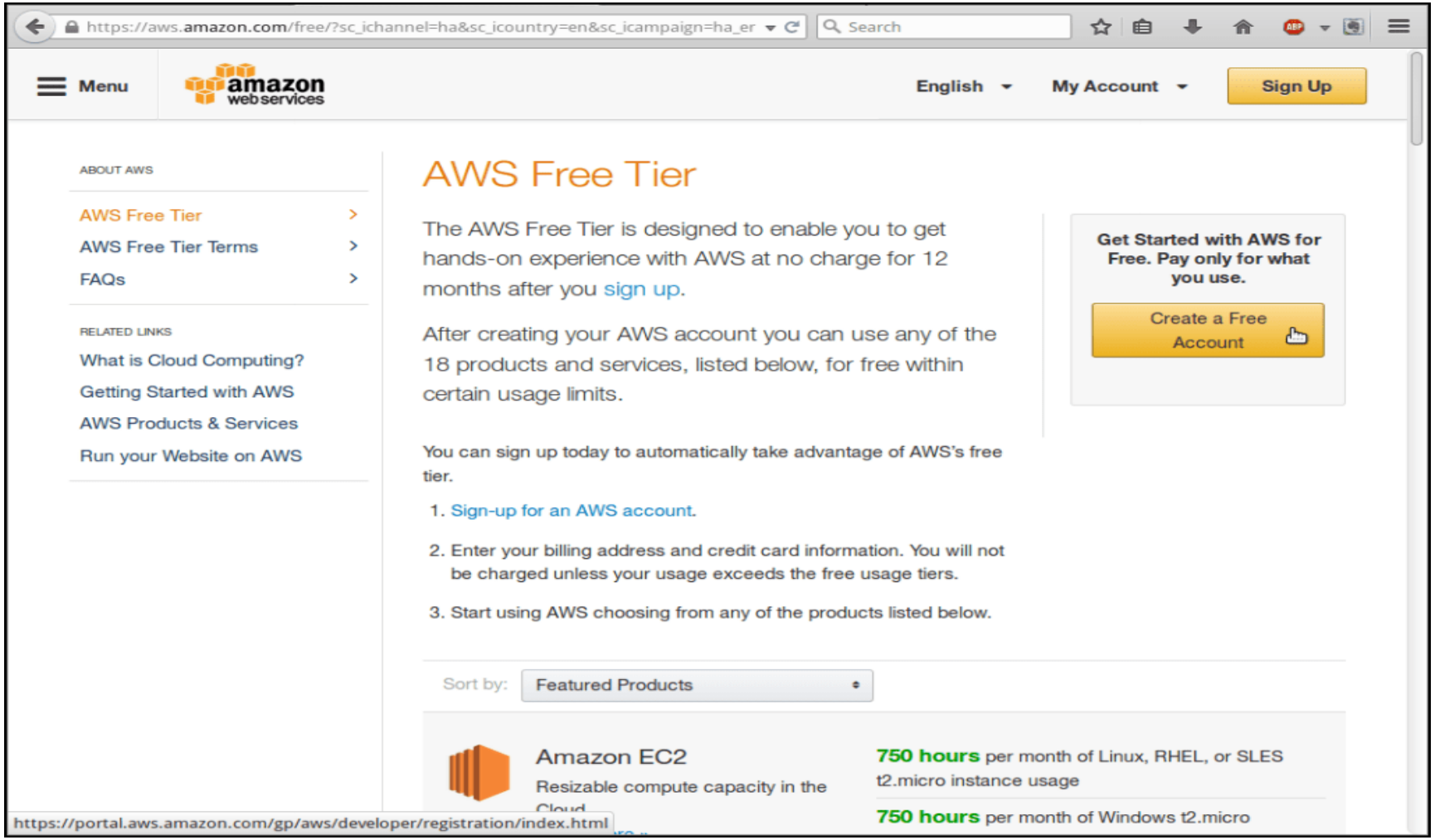
Salaries may vary with your experience in the field

Job Role	Salary in USD
AWS SysOps Administrator	$111,000 - $160,000
Cloud Developer	$95,000
AWS Solutions Architect	$98,000 - $150,000
Cloud Software Engineer	$63,000 - $93,000

AWS Cloud Use Cases

- AWS enables you to build sophisticated, scalable applications
- Applicable to a diverse set of industries
- Use cases include
  - Enterprise IT, Backup & Storage, Big Data analytics
  - Website hosting, Mobile & Social Apps
  - Gaming

Service Domains in AWS

- Compute
- Database
- Storage
- Migration & Transfer
- Customer Engagement
- Security, Identity, & Compliance
- Management & Governance
- Networking & Content Delivery

Create an AWS account & my budget plan